

Module Summary: AI Programming Foundations

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Overview

This project builds a reproducible data workflow that reads, cleans, analyzes, and visualizes Boston Marathon finishing data from the 2016 and 2017 editions. The dataset was originally scraped from the Boston Marathon website using a Python script I wrote, and later uploaded to Kaggle, where it has been downloaded over 8,000 times and viewed more than 68,000 times.

Dataset Description

The dataset consists of two CSV files, one per race year, each containing approximately 26,000 runner records. Key variables include runner demographics (name, age, gender, city, state, country), checkpoint split times (5K through 40K), half-marathon time, overall pace, projected time, official finish time, and ranking positions (overall, gender, division). After combining both years, the final dataset contains roughly 53,000 rows and 23 usable columns.

Workflow Description

The workflow follows five stages:

1. Ingestion -- Both CSV files were loaded using Pandas and combined into a single DataFrame. A Year column was added to distinguish between the two editions.
2. Cleaning -- Two cleaning functions were applied. The first removed four columns that were either CSV export artifacts or too sparse to be meaningful (Unnamed: 0, Unnamed: 8, Unnamed: 9, and Citizen, which was 95% null). The second converted all time-string columns from "H:MM:SS" format into total minutes as floats, handling placeholder values ("-") by converting them to NaN.
3. Exploratory Analysis -- A reusable summary function was built to compute count, mean, median, standard deviation, min, and max for any numeric column grouped by a categorical variable. This function was applied to compare finish times by gender, year, and country.
4. Visualizations -- Four figures were created: a histogram comparing finish time distributions across years, a box plot comparing finish times by gender and year, a line plot of median finish time by age, and a personalized overlay showing my own finish times against the overall field distribution.
5. Summary -- Key findings, limitations, and assumptions were documented in the notebook.

Key Decisions and Assumptions

The decision to drop the Citizen column was based on its extreme sparsity -- with only 5% of rows populated, it could not support reliable analysis. Time columns were converted to total minutes rather than kept as timedelta objects because minutes are easier to interpret in plots and summary statistics. The median was chosen over the mean as the primary measure of central tendency because marathon finish times tend to be right-skewed, making the median more representative of a typical runner's performance (Danchev, 2022).

Results and Interpretation

The finish time distributions for both years are remarkably similar in shape (Figure 1), centered between 200 and 250 minutes. This consistency reflects the qualifying standards imposed by the Boston Athletic Association. Male runners show a lower median finish time than female runners (Figure 2), consistent with gender-differentiated qualifying thresholds. Median finish time by age follows a U-shaped curve (Figure 3), with the fastest times in the late 20s to mid 30s and a gradual increase after age 40. As a participant in both editions, I plotted my own results against the field (Figure 4), showing a roughly 40-minute difference between my 2016 and 2017 finishes.

Responsible Practice (Bias and Data Quality)

It is important not to assume that data is free of bias. When analyzing results by country, some nations appear to have dramatically faster median times. However, this reflects selection bias rather than a true national comparison -- countries like Kenya and Ethiopia are represented almost exclusively by elite, sponsored athletes with very small sample sizes, while the United States contributes tens of thousands of runners spanning the full ability spectrum. Comparing these medians as if they are equivalent would be misleading. As Van den Broeck et al. (2005) note, data anomalies can lead to biased parameter estimates and flawed conclusions if not properly identified and addressed during the cleaning process. This dataset also does not account for variability in race-day conditions (weather, wind, temperature), which could affect overall results and disproportionately impact average runners more than elites.

Reproducibility

The project is fully reproducible. A requirements.txt file was generated using pip freeze, capturing all dependencies. The GitHub repository contains multiple commits showing incremental progress and at least one development branch beyond main. The README includes instructions for cloning the repository, installing dependencies, and running the notebook. The dataset CSV files are included in the repository so that the notebook can be executed without external downloads.

References

- Danchev, V. (2022). Reproducible Data Science with Python: An Open Learning Resource. *Journal of Open Source Education*, 5, 156. <https://doi.org/10.21105/jose.00156>
- Van den Broeck, J., Cunningham, S. A., Eeckels, R., & Herbst, K. (2005). Data Cleaning: Detecting, Diagnosing, and Editing Data Abnormalities. *PLoS Medicine*, 2(10), e267. <https://doi.org/10.1371/journal.pmed.0020267>